1. Introduction to Data Cleaning / Data Preprocessing

Definition

Data Cleaning (also called **Data Preprocessing**) is the process of preparing raw data before using it in Machine Learning or Data Analysis.

It involves:

- finding errors
- fixing missing values
- converting text into numeric form
- removing duplicates
- handling outliers
- scaling features
- correcting data types

Simple Meaning

Raw data is usually **messy**.

It may contain:

- blank values
- wrong formats
- repeated rows
- text where numbers are needed
- very large/small abnormal values
- categories in text form

Before giving this data to a machine learning model, we must **clean and transform it**.

That process is called:

Data Cleaning / Data Preprocessing

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2. Why Data Cleaning is Important in Machine Learning

This is the most important point of the lecture.

Core Idea

Machine Learning models do not understand "dirty data."
They only learn from the data we provide.
If the data is bad, the model will also learn badly.

Famous Rule

Garbage In = Garbage Out (GIGO)

If input data is poor quality:

- predictions become wrong

- accuracy becomes low
- model becomes biased
- training becomes unstable
- real-world decisions become unreliable

Real-Life Example

Imagine a hospital wants to predict diabetes risk.

If data has problems like:

- missing glucose values
- age stored as text
- duplicate patient records
- extreme wrong values like glucose = 9999
- gender as text not encoded

Then the ML model may give **wrong medical predictions**.

That is dangerous.

So data cleaning is not optional — it is **essential**.

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3. Where Data Cleaning is Used in Data Science

Data cleaning is used in **every single data science project**.

Before:

- EDA (Exploratory Data Analysis)
- Statistical analysis
- Machine learning
- Deep learning
- Dashboard reporting
- Business intelligence

Domains:

- Healthcare
- Banking
- HR analytics
- E-commerce
- Education
- Fraud detection
- Customer churn prediction

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4. Full Data Cleaning Pipeline (Flow of Today's Topic)

A good flow to teach is:

1. Identifying Missing Values
2. Handling Missing Values
3. Encoding Categorical Variables
    - One Hot Encoding
    - Dummy Variables
    - Label Encoding
    - Ordinal Encoding
4. Outlier Detection and Removal
5. Feature Scaling
    - Standardization
    - Normalization
6. Handling Duplicate Data
7. Data Type Transformation

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5. Identifying and Handling Missing Values

Definition

A **missing value** is a data value that is not available or not recorded.

It may appear as:

- NaN
- None
- blank cell
- null
- empty string
- special symbols like ?

Real-Life Example

A student performance dataset:

Name	Attendance	StudyHours	Marks
A	90	4	78
B	NaN	5	82
C	85	NaN	75

Here:

- Attendance is missing for B
- StudyHours is missing for C

Why Missing Values Matter

If missing values are not handled:

- many ML algorithms will throw errors
- statistics like mean/median become incorrect
- model can become biased
- predictions may become unstable

How to Identify Missing Values

We check:

- total missing values in each column
- percentage of missing data
- whether missingness is random or systematic

Ways to Handle Missing Values

1. Remove Rows

Use when:

- only a few rows are missing
- dataset is large
- those rows are not important

Example:

If 2 rows out of 10,000 are missing → remove them.

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2. Remove Columns

Use when:

- a column has too many missing values
- column is not important
- column has little business value

Example:

If a column has 80% missing values, it may be dropped.

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3. Fill with Mean

Used for **numerical continuous data**.

Example:
If missing values in Salary, replace with average salary.

Good when data is fairly symmetric.

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4. Fill with Median

Used for numerical data when:

- data has outliers
- distribution is skewed

Example:
House prices often use median instead of mean.

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5. Fill with Mode

Used for **categorical data**.

Example:
Missing values in City, Gender, Department

Replace with most frequent category.

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6. Forward Fill / Backward Fill

Used mainly in:

- time series data
- sequential records

Example:
Stock prices, sensor data, daily temperature

Data Science Importance

Handling missing values correctly improves:

- model accuracy
- fairness
- data consistency
- stability of training

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6. One Hot Encoding & Dummy Variables

This is one of the most important preprocessing topics.

Why Encoding is Needed

Machine learning models generally work with **numbers**, not raw text.

If a column contains:

- Male / Female
- Red / Blue / Green
- Delhi / Mumbai / Ahmedabad

The model cannot directly understand these text values.

So we convert them into numerical form.

A. One Hot Encoding

Definition

One Hot Encoding converts each category into a separate binary column (0 or 1).

Example

Column: Color

Color
Red
Blue
Green

After One Hot Encoding:

Color_Red	Color_Blue	Color_Green
1	0	0
0	1	0
0	0	1

Meaning

Each category gets its own column.

- If the row belongs to that category → 1
- Otherwise → 0

Why It Is Useful

It prevents the model from thinking categories have an order.

Example:

If we encode Red=1, Blue=2, Green=3 directly, the model may think:

Green > Blue > Red

But color has **no natural order**.

So One Hot Encoding is better.

Where Used

Used for **nominal categorical variables**, such as:

- city
- gender
- product type
- payment mode
- browser type

B. Dummy Variables

Definition

Dummy variables are similar to One Hot Encoding, but we drop **one category column** to avoid redundancy.

This is often called:

k - 1 encoding

If there are k categories, we create k - 1 columns.

Example

For Color = Red, Blue, Green

Instead of 3 columns, create 2:

Color_Red	Color_Blue
1	0
0	1
0	0

Why Dummy Variables Are Used

To avoid:

Dummy Variable Trap

This happens when one variable can be perfectly predicted from others, causing **multicollinearity**.

Especially important in:

- linear regression
- logistic regression

Difference: One Hot vs Dummy Variables

- **One Hot Encoding** → creates all category columns

- **Dummy Variables** → drops one column

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7. Label Encoding

Definition

Label Encoding assigns a unique integer to each category.

Example

Column: City

City
Delhi
Mumbai
Ahmedabad

After Label Encoding:

City
0
1
2

Important Warning

This may create a **false order**.

The model may think:

Ahmedabad < Delhi < Mumbai

But cities have no natural ranking.

So label encoding is **not ideal for nominal categories**.

Best Use Case

Use Label Encoding when:

- target/output variable is categorical
- tree-based models may handle it
- categories are many and simple
- order is not harmful in specific context

Where Used in Data Science

- target variable encoding

- binary categories
- quick preprocessing for tree models
- intermediate transformation steps

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8. Ordinal Encoding

Definition

Ordinal Encoding is used when categories have a **natural order or ranking**.

Example

Column: Education

Education
High School
Bachelor
Master
PhD

After Ordinal Encoding:

Education
1
2
3
4

Why This Is Correct

Because here there **is a meaningful order**:

High School < Bachelor < Master < PhD

So assigning numbers makes sense.

More Real-Life Examples

- Customer satisfaction: Low, Medium, High
- T-shirt size: Small, Medium, Large
- Risk level: Low, Moderate, High, Critical
- Rating: Poor, Average, Good, Excellent

Difference: Label Encoding vs Ordinal Encoding

- **Label Encoding** → numbers assigned, but no meaningful order guaranteed
- **Ordinal Encoding** → numbers assigned **based on real order**

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9. Outlier Detection and Removal

This is a very important statistical + ML topic.

Definition

An **outlier** is a data point that is **very different from the rest of the data**.

It is unusually high or unusually low compared to normal values.

Real-Life Example

Salary data:

25000, 28000, 30000, 32000, 29000, 31000, 500000

Here:

- 500000 is much larger than the rest

So it is an **outlier**.

Why Outliers Matter

Outliers can:

- distort mean
- affect standard deviation
- skew distribution
- mislead statistical analysis
- badly affect regression models
- reduce model accuracy

Where Outliers Can Be Normal

Sometimes outliers are **real and meaningful**.

Example:

- fraud transaction detection
- luxury customer spending
- disease anomalies
- rare but important events

So we should not remove every outlier blindly.

Methods to Detect Outliers

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1. IQR Method (Interquartile Range)

Formula:

- IQR = Q3 - Q1

Outlier limits:

- **Lower Bound = Q1 - 1.5 × IQR**
- **Upper Bound = Q3 + 1.5 × IQR**

Any value outside these bounds is considered an outlier.

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2. Z-Score Method

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X= data value
- μ= mean
- σ= standard deviation

If:

- |Z| > 3 → often considered outlier

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3. Box Plot (Visual Method)

A box plot visually shows:

- median
- quartiles
- spread
- outliers as separate points

How to Handle Outliers

- remove if clearly wrong
- cap (winsorize) extreme values
- transform data (log transform)
- use robust statistics (median, IQR)
- keep them if they are meaningful

Data Science Use

Very important in:

- regression
- financial data
- healthcare analytics
- fraud detection
- sales analysis

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10. Feature Scaling

Feature scaling is critical before many ML algorithms.

Definition

Feature Scaling means adjusting numerical features so they are on a similar scale.

Why Scaling is Needed

Suppose we have:

- Age = 22
- Salary = 750000

Salary is much larger in magnitude.

If scaling is not done:

- some algorithms may give too much importance to Salary
- distance-based algorithms become biased

Algorithms That Need Scaling

Very important for:

- KNN
- K-Means
- SVM
- Logistic Regression
- Linear Regression (often beneficial)
- PCA
- Gradient-based algorithms

Algorithms Less Sensitive to Scaling

Usually not required for:

- Decision Tree
- Random Forest
- XGBoost (less sensitive)

A. Standardization

Definition

Standardization transforms data so that:

- mean becomes **0**
- standard deviation becomes **1**

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X= original value
- μ= mean
- σ= standard deviation

Meaning

It tells how far a value is from the mean in standard deviation units.

Best Use Case

Use when:

- data is approximately normally distributed
- algorithms depend on distance or gradients
- outliers are not extremely severe

B. Normalization (Min-Max Scaling)

Definition

Normalization scales values into a fixed range, usually:

0 to 1

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Meaning

- smallest value becomes 0
- largest value becomes 1
- all other values fall between 0 and 1

Best Use Case

Use when:

- exact bounded range is needed
- neural networks often benefit
- image pixel scaling
- data does not need centered mean

Difference: Standardization vs Normalization

- **Standardization** → mean = 0, std = 1
- **Normalization** → range usually 0 to 1

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11. Handling Duplicate Data

Definition

Duplicate data means **same record appears more than once**.

Example

ID	Name	Marks
1	A	78
2	B	85
2	B	85

Here the second and third rows are duplicates.

Why Duplicates Are a Problem

Duplicates can:

- bias analysis
- make frequency counts wrong
- inflate sample size
- distort model learning
- over-represent some cases

Real-Life Example

In customer churn data, if the same customer appears multiple times incorrectly:

- the model may think that customer behavior is more common than it really is

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How to Handle

- identify full duplicates
- identify partial duplicates based on key columns
- remove only true duplicates
- keep repeated records if they represent real repeated events

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12. Data Type Transformation

This is very practical and important.

Definition

Data Type Transformation means converting columns into the correct data type.

Common types:

- integer (int)
- decimal (float)
- string (object)
- boolean (True/False)
- datetime (date/time)
- category

Why It Is Important

Wrong data types cause:

- incorrect calculations
- errors in model training
- wrong sorting/filtering
- poor memory usage
- incorrect statistical operations

Real-Life Example 1

Column Age stored as text:

"21", "22", "23"

Looks numeric, but it is actually a string.

Problems:

- mean cannot be calculated properly
- scaling won't work
- ML model cannot use it correctly

So it should be converted to numeric.

Real-Life Example 2

Column Joining_Date stored as text:

"12-01-2024"

If not converted to datetime:

- cannot extract month
- cannot calculate tenure
- cannot sort by actual date properly

Common Transformations

- string → integer
- string → float
- string → datetime
- object → category
- boolean conversion
- numeric coercion with error handling

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13. Full Concept Flow (Best Understanding Summary)

Here is the simplest overall understanding:

Raw data may contain:

- missing values
- text categories
- duplicates
- wrong types
- abnormal values
- different scales

So before ML:

We must:

- clean missing values

- encode categories
- handle outliers
- scale numeric features
- remove duplicates
- fix data types

Only then data becomes:

Machine Learning Ready

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14. Real-Life End-to-End Example (Very Strong for Class)

Suppose we have an **employee salary prediction dataset**:

Columns:

- Age
- Education
- Experience
- City
- Salary

Problems:

- Age has missing values
- Education is text
- City is text
- Some salary values are extremely high
- Duplicate employee records exist
- Experience is stored as text

Preprocessing Steps:

1. Fill missing Age with median
2. Encode Education using Ordinal Encoding
3. Encode City using One Hot Encoding
4. Detect salary outliers using IQR
5. Remove duplicate records
6. Convert Experience to numeric
7. Scale Age and Experience

Now the dataset is ready for ML.

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15. Why This Topic Is Critical Before Algorithms

This is a strong line you can say in class:

A good machine learning model does not start with an algorithm.
It starts with clean, meaningful, and properly prepared data.

Another strong line:

In real-world data science projects, data preprocessing often takes more time than model building itself.

This is 100% true in industry.

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16. Strong Interview / Viva Important Points

Students should remember:

- Missing values can break models
- Encoding is needed because ML needs numeric input
- One Hot is for nominal categories
- Ordinal Encoding is for ordered categories
- Outliers can distort analysis
- Scaling is essential for distance-based models
- Duplicates can bias data
- Correct data types are necessary for calculations and ML

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17. Final Summary

Data Cleaning / Data Preprocessing is the process of converting raw, messy, inconsistent data into clean, structured, machine-learning-ready data.

It includes:

- handling missing values
- encoding categorical variables
- removing or treating outliers
- scaling features
- removing duplicate rows
- transforming data types

This step is critical because:

The quality of a machine learning model depends heavily on the quality of the input data.

In machine learning, better data usually gives better results than a more complicated algorithm.